Zhi Hong

Email | Google Scholar | GitHub

Education

South China University of Technology (SCUT) Guangzhou, China
Dual-Degree Program Sep 2022 – Jun 2026
B.Eng. in Automation (Control Science and Engineering)
B.Sc. in Mathematics and Applied Mathematics
GPA: 3.84/4.0 **Rank:** 3/24

Relevant Coursework: Mathematical Analysis, Advanced Algebra, Probability Theory, Mathematical Statistics, Stochastic Processes, Principles of Automatic Control, Machine Learning, Deep Learning, Reinforcement Learning

Research Interests

Large Language Models: Harness Engineering, Multi-Agent Systems, Social Simulation, Post-Training

Machine Learning: Bandits, Reinforcement Learning, Federated Learning

My research focuses on the design and optimization of LLM-based agent systems. At the single-agent level, I study harness engineering, focusing on how agents reason, retrieve relevant memories, and evolve by learning from past experience. At the multi-agent level, I investigate how to automatically build MAS and improve coordination among agents to enhance collective performance. I also explore agent-based social simulation, both to understand agent interactions and collective behavior and to generate data for post-training.

Publications

1. **MASPOB: Bandit-Based Prompt Optimization for Multi-Agent Systems with Graph Neural Networks**
   Zhi Hong[*], Qian Zhang[*], Jiahang Sun, Zhiwei Shang, Mingze Kong, Xiangyi Wang, Yao Shu[†], Zhongxiang Dai[†].
   ICML 2026 **(Spotlight, Top 2.2% of 23,918 submissions)**

Preprints & Under Review

1. **HARNESSFLOW: MCTS-Guided Harness Optimization for LLM Agents**
   Zhi Hong, Qianlong Wang, Hongchen Wei, Aidan Wu, Daohong Ying, Mingze Kong, Bei Liu[†], Chong Luo.
   arXiv:2611.06254
2. **OfficeTown: Generative Organizations for Evolving Agent Intelligence**
   Hongchen Wei[*], Daohong Ying[*], **Zhi Hong**[*], Aidan Wu[*], Bei Liu[†], Ruichun Ma, Yifan Yang, Qi Dai, Kai Qiu, Rui Xu, Dongdong Chen, Zhenzhong Chen, Chong Luo. arXiv:2610.23194, Project Page
3. **Which LLM for Which Agent? Topology-Aware Bandit-based Model Selection for Multi-Agent Systems**
   Zhi Hong, Jie Mao, Qianlong Wang, Daohong Ying, Shuang Qiu, Bei Liu, Zhongxiang Dai[†].
4. **Cost-Aware Multi-Objective Bandits: Theory and Application to Budgeted LLM Configuration Evaluation**
   Bo Xue, **Zhi Hong**, Jiayi Li, Yuanyu Wan, Ji Cheng, Shuang Qiu. arXiv:2608.04333
5. **FedPOB: Sample-Efficient Federated Prompt Optimization via Bandits**
   Pingchen Lu[*], **Zhi Hong**[*], Zhiwei Shang, Zhiyong Wang, Yikun Ban, Yao Shu, Min Zhang, Shuang Qiu, Zhongxiang Dai[†]. arXiv:2509.24701
6. **AgentSpec: Understanding Embodied Agent Scaffolds Through Controlled Composition**
   Jixuan Chen, Jianzhi Shen, Haoqiang Kang, **Zhi Hong**, Qingyi Jiang, Soham Bose, Yiming Zhang, Leon Leng, Amit Vyas, Lingjun Mao, Siru Ouyang, Kun Zhou, Lianhui Qin. arXiv:2606.14674
7. **Entropy Forcing: Entropy-Routed Subset Selection for KV Cache in Streaming Video Generation**
   Qianlong Wang, Zhikang Xie, Xichen Ye, **Zhi Hong**, Wenzhe Cao, Yifan Wu, Cheng Jin, Weizhong Zhang, Zengfeng Huang.
8. **Workflow-R1: Group Sub-sequence Policy Optimization for Multi-turn Workflow Construction**
   Mingze Kong, Zikun Qu, Zhongquan Zhou, Pengyu Liang, Xiang Li, Zhiwei Shang, **Zhi Hong**, Kaiyu Huang, Zhiyong Wang, Zhongxiang Dai[†]. arXiv:2602.01202
9. **Linear and Neural Dueling Bandits with Delayed Feedback**
   Xiangyi Wang, Pingchen Lu, Jie Mao, Mingze Kong, **Zhi Hong**, Zhiyong Wang, Zhongxiang Dai. arXiv:2605.26554

Research Experience

Microsoft Research Asia (MSRA) Jun 2026 – Present

Research Intern (Star Bridge Program), Visual Computing Group
Advisor: Bei Liu

- Investigated automated harness optimization for LLM agents, using execution feedback to guide code-level refinement and the reuse of agent components.
- Examined how harness design affects task performance, inference cost, and search efficiency across agentic, coding, and web-interaction benchmarks.
- Contributed to organizational multi-agent simulation for analyzing collaborative behavior and generating interaction data for agent learning and post-training.

University of California San Diego Jan 2026 – May 2026

Research Intern, Computer Science and Engineering
Advisor: Prof. Lianhui Qin; Mentor: Jixuan Chen

- Contributed to the design and integration of reasoning, memory, and reflection mechanisms in modular LLM-based embodied agents.
- Conducted controlled experiments to assess how component composition and interactions affect agent performance.

The Chinese University of Hong Kong, Shenzhen Sep 2024 – Jul 2026

Research Assistant, School of Data Science
Advisor: Prof. Zhongxiang Dai

- Investigated bandit-based prompt optimization for multi-agent systems, using graph-based representations to model inter-agent dependencies under limited evaluation budgets.
- Studied topology-aware model selection for multi-agent systems, modeling inter-agent dependencies to improve the trade-off between task performance and inference cost.
- Contributed to reinforcement learning for workflow construction, aligning policy optimization with the reasoning–action structure of multi-turn agent interactions.
- Studied cost-aware multi-objective bandits for budgeted LLM configuration evaluation, including online selection and fixed-budget Pareto identification.
- Investigated federated prompt optimization using bandit algorithms to improve sample efficiency across distributed clients.

Honors & Awards

Outstanding Graduate – South China University of Technology 2026

First Prize – China Mathematics Competition (CMC, Category A, Province Level) 2024

National Encouragement Scholarship – Ministry of Education, China 2023 – 2025

Merit Student – South China University of Technology 2023 – 2025

Skills & Interests

Languages: English (TOEFL iBT 5.0/6.0), Chinese (Native)

Programming: Python, LaTeX, C++, MATLAB

Research Skills: Reinforcement Learning, Deep Learning, Post Training, Bandit Algorithms

Interests: Traveling, Basketball, Swimming

Academic Service

Reviewer: ICLR (2027)